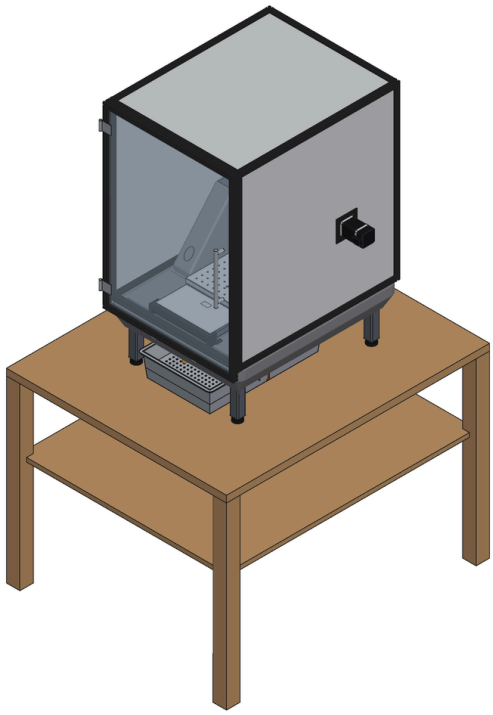
# Build guide 5: enclosure, coolant and commissioning

DIY CNC epoxy-granite mill · October 2026

## 8. Add-ons, enclosure and coolant (the add-later items and part 3 of the sourcing list)

The machine runs without these: until then, brush or drip cutting fluid onto the cut and let the gutters and drain carry it to a bin under the base. Everything here bolts to fixings that are already in the castings, so nothing is ever recast: fit them in any order, as you buy them. Fit the way covers before steady steel work; chips wear the Y screw and rails.

1. Bolt the 2020 frame to the base top with angle brackets into the cast-in M5 inserts (M5 x 35). Cut list in `docs/5-design/report.txt`.
2. ACM side, back and roof panels; polycarbonate door on two hinges on the front left post, so it swings open to the left (away from the X motor) and stays put without struts; magnetic catch. Brush seals where the X and Y motors pass through the panels.
3. Way covers: the stainless strips on their spring rollers in front of and behind the table. Each roller's feet screw down (M5 x 12): the front roller's into two inserts behind the coolant trough, the rear rollers' outer feet into two inserts and their inner feet into the tapped pad beside the Y bracket. The spiral cover clamps on the Y screw between the nut housing and the bracket's bearing end. The drip shield (ACM offcut) goes over the X rails, its back edge under the washers of the four front beam studs: back the nuts off, slide it in, retighten.
4. Coolant: slide the sump (steam-table pan with the chip basket, magnet, baffle, screen and 24 V pump) in under the base below the drain nipple, in place of the bin. Run the return hose to the bulkhead, up the cable chain, to the manifold and nozzles on the spindle clamp. Use water-soluble coolant.



## 9. Commissioning

1. **Run each axis by hand,** then with the motors at low speed, end to end. Listen for tight spots.
2. **Homing and soft limits** in FluidNC; check the travel (180 / 300 / 140 mm).
3. **Steps per mm:** move 100 mm and measure with the dial indicator or a caliper; correct.
4. **Squareness:** sweep an indicator in the spindle along a straightedge or the table edge. X to Y is set by the beam position (step 4 of the frame); Z to the table is set by shimming the spindle clamp (tram). Aim for under 0.02 mm over 100 mm.
5. **Face the table:** take a light fly-cut over the aluminum table so it is flat and parallel to X/Y.
6. **First cuts:** aluminum first, then steel at light depths of cut with flood coolant. Check the spindle temperature and the coolant flow, and re-torque the frame bolts and the riser hold-downs after the first few hours of running.